

Written examination in

Dynamic Models of Electrical Machines and Control Systems

1st semester M.Sc. (PED/EP SH/WPS/MCE)

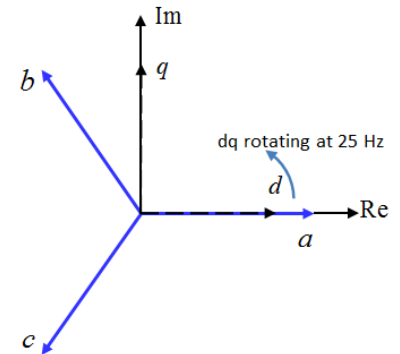
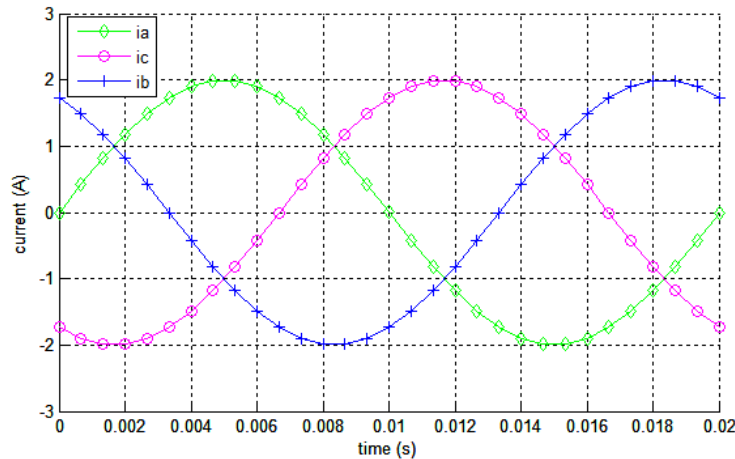
Duration: 4 hours
(Reviewed by Dong Wang)

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- All usual helping aids are allowed, including text books, slides, personal notes, and exercise solutions
 - Calculators and laptop computers are allowed, provided all wireless and wired communication equipment is turned off
 - Internet access is strictly forbidden
 - Any kind of communication with other students is not allowed
 - Remember to write your study number on all answer sheets
 - All intermediate steps and calculations should be included in your answer sheets --- printing the final result is insufficient
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The set consists of 4 problems

Problem 1 (25%)

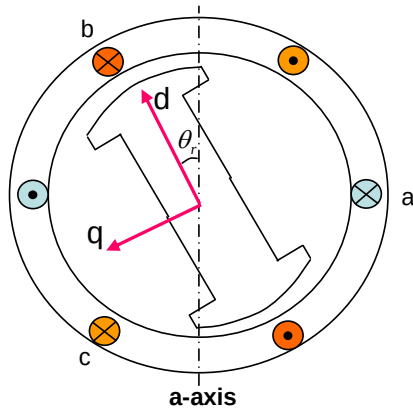
- (1) Observe the following instantaneous a, b and c current waveforms (**Please pay attention to the phase sequence!**)



- Please indicate the rotating direction of the current space vector (clockwise or anti-clockwise)? Please explain.
 - Please show the current space vector with respect to phase-a axis at time $t = 0.0075$ (seconds)
 - Please draw the corresponding $\alpha\beta$ components for the time interval of $[0.01, 0.02]$ seconds.
 - A rotating dq reference frame is chosen. Its rotating frequency is at 100 Hz, anticlockwise direction. At $t = 0$, its d-axis is aligned with phase a-axis (as indicated above). Please draw the corresponding dq components for the time interval of $[0, 0.02]$ seconds.
- (2) Please find the space vector of the following a, b, c signals
 $(v_a = V_{pk} \cos(\omega_e t), v_b = V_{pk} \cos(\omega_e t + \frac{2\pi}{3}), v_c = V_{pk} \cos(\omega_e t - \frac{2\pi}{3}))$, using the same a, b, c-axes as shown in question (1).
 Then, give the corresponding α and β components (viewed from a $\alpha\beta$ reference frame).

Problem 2 (25%)

A sketch of a synchronous machine is shown below.



- (1) Please show how the mutual inductance between phase-a and phase-c may be derived.
- (2) Please sketch this inductance vs. rotor position waveform (Y-axis: the above mutual inductance; X-axis: rotor position).

A simple single-phase PM machine is shown below in Fig. 2(a). Now another phase (naming it phase-b) is added to this machine, as indicated in Fig. 2(b).

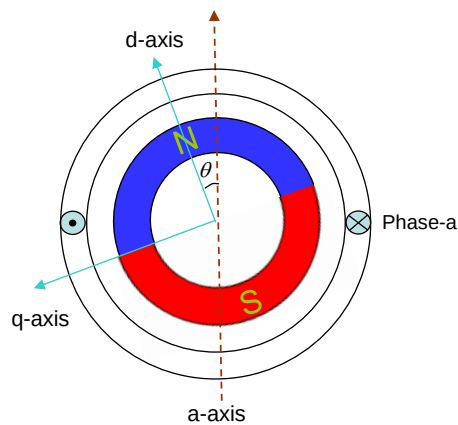


Fig. 2(a)

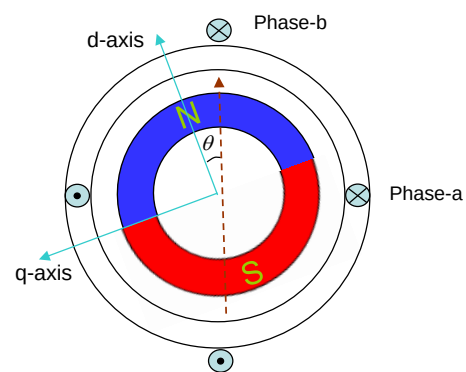


Fig. 2(b)

- (3) If the phase-a PM flux linkage waveform is expressed as: $\lambda_{pm,a} = \lambda_{mpm} \cos \theta$. (where λ_{mpm} is its peak value and θ is the rotor position as indicated in Fig. 2), please give the PM flux linkage waveform for phase-b.
- (4) If phase-a is now supplied with a current of $i_a = -I_m \sin \theta$ (where I_m is the peak value of the current), please determine the needed current waveform for phase-b, so that phase-b can produce the same torque profile as phase-a.
- (5) Please show the instantaneous torque produced by phase-a and phase-b, respectively.
- (6) Please give an expression for the total torque produced by phase-a and phase-b together.

Problem 3 (25 %)

An induction motor has the following data (**the rotor windings are short-circuited; the machine is Y-connected**):

Rated shaft power	7.5 kW
Rated speed	1160 rpm
Rated stator frequency	60 Hz
Number of poles	6
Rated stator voltage	380 V RMS (line-to-line)
Rated phase current	14 (A) RMS
Rated power factor $\cos \phi$	0.8 inductive
Stator resistance	0.28 Ohm

The stator side voltage equation may be expressed as

$$\begin{bmatrix} u_{qs} \\ u_{ds} \\ u_{0s} \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \cdot \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{0s} \end{bmatrix} + p \begin{bmatrix} \lambda_{qs} \\ \lambda_{ds} \\ \lambda_{0s} \end{bmatrix} - \omega_\theta \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \lambda_{qs} \\ \lambda_{ds} \\ \lambda_{0s} \end{bmatrix}$$

- (1) Please give the vector form of this voltage equation (taking q as the real component and d as the minus imaginary component, i.e. qd-reference frame).
- (2) Please give the stator side voltage equation in a vector form expressed in the $\alpha\beta$ -reference frame. Please calculate the magnitude of the stator flux linkage ($|\bar{\lambda}_{\alpha\beta s}|$) at the rated steady state operation condition.
- (3) Now the motor is running at 0.1 Hz (electrical frequency) under V/f control. In order to maintain the same stator flux level as experienced at the rated condition, compensation of the voltage drop on the stator resistance needs to be introduced to the V/f control. In steady state, the phase-b current peak value is found to be 1.0 (A) and is lagging phase-b voltage by 30 degrees. Please determine the magnitude of the phase voltage vector after phase resistive voltage drop compensation.
- (4) In V/f control, the stator frequency is now 50% of the rated frequency. The load torque is 25% of the rated torque. What is the slip frequency in Hz that needs to be compensated in order to make the **mechanical** rotor shaft speed to be 10 Hz?

Problem 4 (25 %)

The stator voltage equation of a permanent magnet synchronous machine may be given as (*same notations as used in the lecture slides*):

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

The number of pole pairs = 4

- (1) This PM machine is driven by another DC motor and running at a constant speed of 1200 rpm. When the stator windings are open-circuited, measured line-to-line RMS voltage is 120 V. Please determine the value of λ_{mpm} to be used in the above machine equations; please also determine the corresponding d-, q-axes open-circuit voltages in dq0-reference frame.
- (2) Please draw the block diagram indicating how you will implement the d-axis voltage and flux linkage equations in Simulink; what is the initial value to be set in the integrator in the implemented Simulink block diagram?
- (3) For this PM machine, it is found that $L_d > L_q$, please sketch a possible location of the current vector with respect to the q-axis for achieving maximum torque per ampere operation.
- (4) At a particular moment ($t = 0$), it is observed that the machine q-axis current is 3 (A) and its d-axis current is 1 (A) (in dq0 reference frame). At this moment, the rotor d-axis is leading the stator phase-a by 30 electrical degrees. The speed is constant and is 1200 rpm. Please draw stator phase-c current waveform for one period, starting from $t = 0$ as defined before. (Please indicate clearly *its initial current value at $t = 0$ and its peak value.*)
The power factor of this machine at this operation condition is 0.866 (voltage leading current). The phase rms voltage is 100 volts. Please add phase-c voltage waveform for one electrical period to the phase-c current waveform obtained previously.

Problem 1 reference frame transformation

It can be observed that: $\theta = 2\pi f \cdot t$, where $f = 50 \text{ Hz}$.

for one period: $2\pi \rightarrow 0.02 \text{ (s)}$

$$\hat{i}_a = 2 \sin \theta; \quad \hat{i}_c = 2 \sin (\theta - 120^\circ); \quad \hat{i}_b = 2 \sin (\theta - 240^\circ)$$

when $\theta = 120^\circ$, $\hat{i}_c = 0$

To have the correct phase sequence:

$$\hat{i}_a = -2 \sin(-\theta) \quad \hat{i}_c = -2 \sin(-\theta + 120^\circ) = -2 \sin(-\theta - 240^\circ);$$

$$\hat{i}_b = -2 \sin(-\theta + 240^\circ) = -2 \sin(-\theta - 120^\circ)$$

Now turn "sin" to "cos"

$$\hat{i}_a = 2 \cos [(-\theta) + 90^\circ] \quad \hat{i}_b = 2 \cos [(-\theta) + 90^\circ - 120^\circ]$$

$$\hat{i}_c = 2 \cos [(-\theta) + 90^\circ - 240^\circ]$$

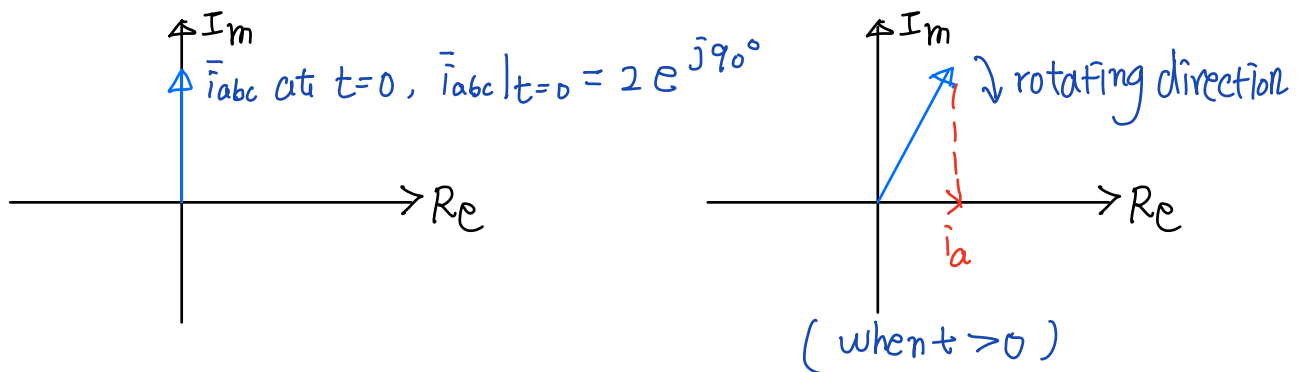
Therefore, according to lecture 2, we have the space vector

$$\bar{i}_{abc} = 2 e^{j[(-\theta) + 90^\circ]}$$

Since $-\theta = -\omega t = (-\omega) \cdot t$ it rotates at the clockwise direction (negative)

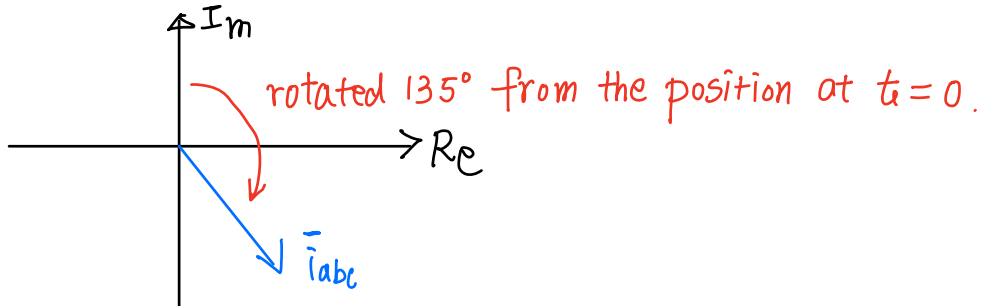
OR you can check: $\hat{i}_a = \operatorname{Re} \left(\frac{\bar{i}_{abc}}{e^{j90^\circ}} \right) = 2 \cos (-\theta + 90^\circ) = 2 \sin \theta$

OR you can think about when $t=0$, $\theta=0$, the space vector is



When $t = 0.0075$ (s). $\theta = 2\pi \times 50 \times 0.0075 = \frac{3\pi}{4} = 135^\circ$

Therefore the location of the current vector: $-135^\circ + 90^\circ = -45^\circ$



For the time interval $[0.01, 0.02]$ seconds, it corresponds to $[\pi, 2\pi]$ region.

$$\bar{i}_a = \bar{i}_a = 2 \sin \theta$$

$$\bar{i}_\beta = \operatorname{Re} \left(\frac{2e^{j(\theta+90^\circ)}}{e^{j90^\circ}} \right) = \operatorname{Re} (2e^{j(\theta-90^\circ)}) = 2 \cos(\theta - 90^\circ) = 2 \cos \theta$$

The waveforms in $[\pi, 2\pi]$ can be found.

- The rotating of the dq can be described as

$$e^{j(2\pi 100 \cdot t)} \quad \text{this is also the location of the d-axis}$$

$$= e^{j(2\theta t)} \quad , \quad \text{where } \theta = 2\pi 50$$

So:

$$\hat{i}_d = \text{Re} \left(\frac{2e^{j(-\theta+90^\circ)}}{e^{j(2\theta)}} \right) = 2 \cos(-3\theta+90^\circ) = 2 \sin(3\theta)$$

$$\hat{i}_q = \text{Re} \left(\frac{2e^{j(-\theta+90^\circ)}}{e^{j(2\theta+90^\circ)}} \right) = 2 \cos(-3\theta) = 2 \cos(3\theta)$$

Its waveform can then be drawn.

- It is a negative sequence as discussed in lecture 2.

So:

$$\bar{V}_{abc} = V_{pk} \cdot e^{-j(\omega t)} \quad \left(\text{You can try to use e.g. } \hat{i}_b = \text{Re} \left(\frac{\bar{V}_{abc}}{e^{j120^\circ}} \right) \text{ to validate} \right).$$

$$V_a = V_{pk} \cos(\omega t)$$

$$V_b = \text{Re} \left(\frac{V_{pk} e^{-j(\omega t)}}{e^{j90^\circ}} \right) = V_{pk} \cos(\omega t + 90^\circ) = -V_{pk} \sin(\omega t)$$

Problem 2, regarding the torque

For phase-a: $\lambda_{pm,a} = \lambda_{mpm} \cos \theta$, when $\theta = 0^\circ$, $\lambda_{pm,a} = \lambda_{mpm}$
(maximum value)

For phase-b: when $\theta = \frac{\pi}{2}$, the rotor N-pole is aligned with phase-b axis
 $\Rightarrow$ max. rotor flux linkage $= \lambda_{mpm}$

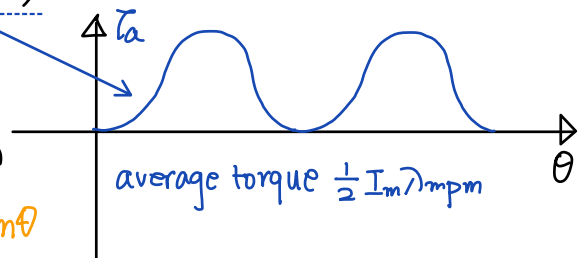
therefore: $\lambda_{pm,b} = \lambda_{mpm} \cos(\theta - \frac{\pi}{2})$

If $\hat{i}_a = -I_m \sin \theta$, then the torque (Note, 1 pole pair, $\omega_{r,e} = \omega_{r,mec}$)

$$\tau_a = \frac{\hat{i}_a \cdot \frac{d\lambda_{pm,a}}{dt}}{\omega_{r,mec}} = \frac{\hat{i}_a \cdot \frac{d\lambda_{pm,a}}{d\theta} \cdot \omega_{r,e}}{\omega_{r,mec}} = \hat{i}_a \cdot \underbrace{\frac{d\lambda_{pm,a}}{d\theta}}_{\text{back-EMF}} = \hat{i}_a \cdot -\lambda_{mpm} \sin \theta$$

$$\tau_a = I_m \lambda_{mpm} \cdot \sin^2 \theta = \frac{1}{2} I_m \lambda_{mpm} (1 - \cos 2\theta)$$

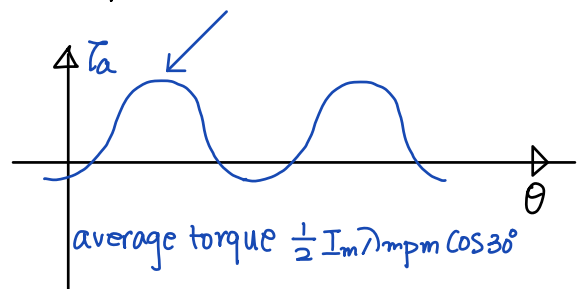
Note, if we DO NOT choose the current to have the same waveform as the back-EMF waveform of $-\sin \theta$



For example: $\hat{i}_a = -I_m \sin(\theta + 30^\circ)$,

$$\tau_a = I_m \lambda_{mpm} \sin \theta \cdot \sin(\theta + 30^\circ) = \frac{1}{2} I_m \lambda_{mpm} (\cos 30^\circ - \cos(2\theta + 30^\circ))$$

It is clear that there will be negative instantaneous torque component and the average torque will be reduced.

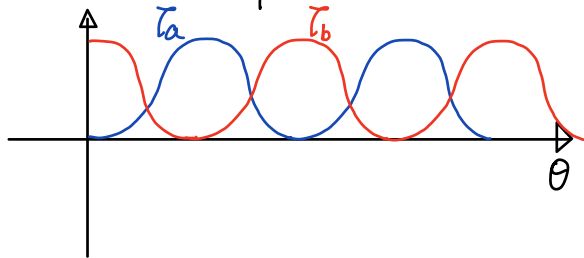


For phase-b : $\lambda_{pm,b} = \lambda_{mpm} \cos(\theta - \frac{\pi}{2})$

then the back-EMF $\frac{d\lambda_{pm,b}}{d\theta} = -\lambda_{mpm} \sin(\theta - \frac{\pi}{2})$

thinking about assuming $\theta' = \theta - \frac{\pi}{2}$, then similar to phase-a
to match the waveform: $i_b = -I_m \sin \theta' = -I_m \sin(\theta - \frac{\pi}{2})$

$$\tau_b = I_m \lambda_{mpm} \sin^2(\theta - \frac{\pi}{2}) = I_m \lambda_{mpm} \cos^2 \theta$$



So the total instantaneous torque:

$$\tau = \tau_a + \tau_b = I_m \cdot \lambda_{mpm}, \text{ which is a const.}$$

Problem 4, regarding the PMSM

① $V_{\text{phase, peak}} = \omega_r \cdot \lambda_{\text{mpm}} = 2\pi \left(\frac{1200}{60} \right) \cdot 4 \cdot \lambda_{\text{mpm}}$ ↖ No. of pole pairs. (should be given)
 ↙ Obtained from the dq-equations with $i_d = i_q = 0$

$\left(\frac{120}{\sqrt{3}} \right) \cdot \sqrt{2}$

phase RMS

it can be found that $\lambda_{\text{mpm}} = 0.195 \text{ (wb)}$

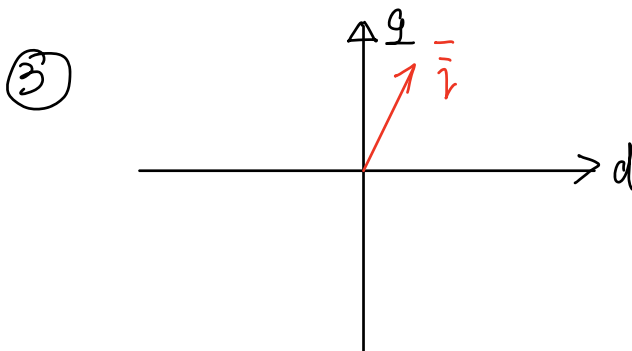
$V_{q, \text{open}} = \omega_r \cdot \lambda_{\text{mpm}} \quad V_{d, \text{open}} = 0$

$V_{\text{phase, peak}} = \sqrt{V_{d, \text{open}}^2 + V_{q, \text{open}}^2} = V_{q, \text{open}}$

② Please refer to the block diagram given on the next page.

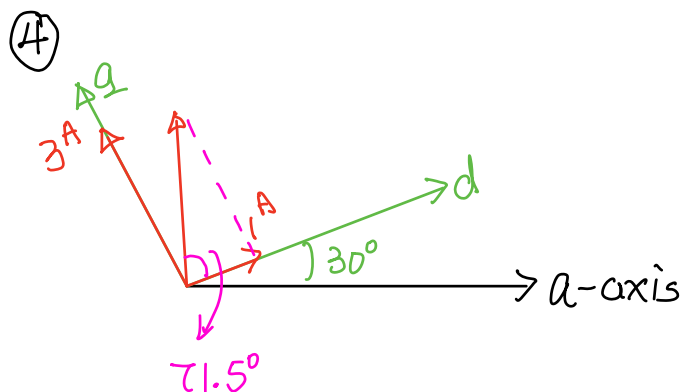
The initial values will be $\lambda_d|_{i_d=0} = \lambda_{\text{mpm}}$

the initial flux linkage values. $\lambda_q|_{i_q=0} = 0$



$i_d > 0$, since the torque e.g.

$T = \frac{3}{2} p \cdot \left[\lambda_{\text{mpm}} \underbrace{i_q}_{>0} + \underbrace{(L_d - L_q)}_{>0} i_d i_q \right]$



$f = \frac{1200}{60} \cdot 4 = 80 \text{ (Hz)}$

one period = 0.0125 (s)

power factor = 0.866 = cos ϕ

$\Rightarrow \phi = 30^\circ$

The current is lagging the voltage by 30°

So. At $t=0$, the location of the current vector is as above: $\bar{i}_{abc} = \sqrt{3^2 + 1^2} \cdot e^{j(101.5^\circ)}$

When $t \uparrow$, the current vector will rotate:

$$\bar{i}_{abc} = \sqrt{3^2 + 1^2} \cdot e^{j(\omega t + 101.5^\circ)}$$

where: $\omega = 2\pi \cdot 80$

The voltage is leading the current by 30° ,

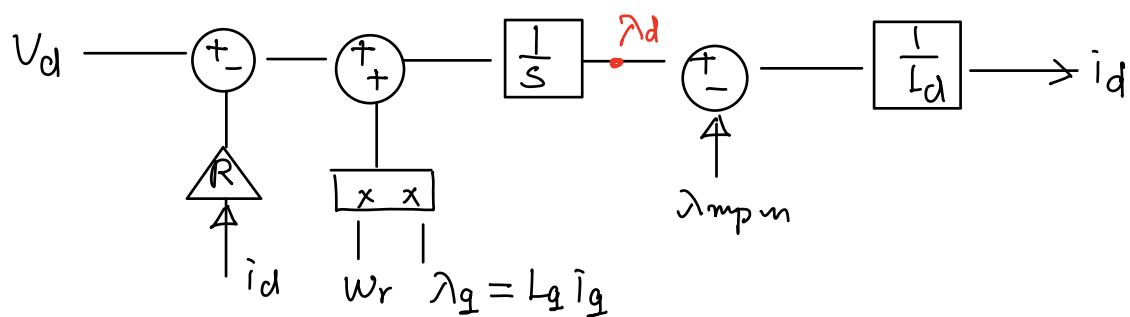
The voltage vector using the phase peak value will be

$$\bar{V}_{abc} = 100\sqrt{2} e^{j(\omega t + 131.5^\circ)}$$

Therefore:

$$V_c = \operatorname{Re} \left(\frac{\bar{V}_{abc}}{e^{j240^\circ}} \right) = 100\sqrt{2} \cos(\omega t - 108.5^\circ)$$

Block diagram for: $V_d = R i_d + \frac{d\lambda_d}{dt} - \omega_r \lambda_q$



Answer to problem 3

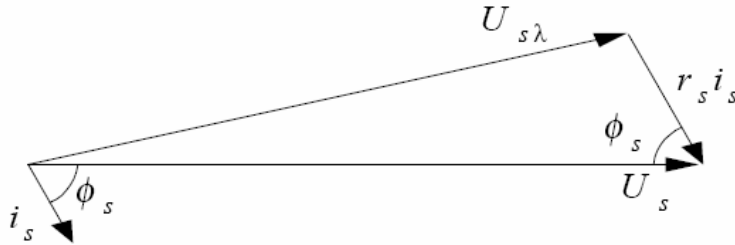
- (1) The vector form of the voltage equation:

$$\begin{aligned}\bar{u}_{qds} &= u_q - j u_d \\ &= (R_s i_{qs} + p \lambda_{qs} + \omega_\theta \lambda_{ds}) - j (R_s i_{ds} + p \lambda_{ds} - \omega_\theta \lambda_{qs}) \\ &= R_s \bar{i}_{qds} + p \bar{\lambda}_{qds} + j \omega_\theta \bar{\lambda}_{qds}\end{aligned}$$

- (2) The stator side voltage equation in a vector form expressed in the $\alpha\beta$ -reference frame:

$$\bar{u}_{\alpha\beta s} = R_s \bar{i}_{\alpha\beta s} + \frac{d \bar{\lambda}_{\alpha\beta s}}{dt}$$

The phasor diagram of the given equivalent circuit can be drawn like below (lecture 11 slide, page 13):



Then, it can be found that:

$$U_{s\lambda}^2 = (U_s - r_s i_s \cos \phi_s)^2 + (r_s i_s \sin \phi_s)^2$$

Thus, it can be calculated that at rated condition ($\phi_s = \varphi$):

$$\begin{aligned}U_{s\lambda} &= \sqrt{(U_s - r_s i_s \cos \varphi)^2 + (r_s i_s \sin \varphi)^2} \\ &= \sqrt{(380/\sqrt{3} - 0.28 \times 14 \times 0.8)^2 + (0.28 \times 14 \times 0.6)^2} = 216.27 \text{ Vrms}\end{aligned}$$

Which is 305.85 V amplitude.

The magnitude of flux linkage:

$$\lambda_s = \frac{\sqrt{2} U_{s\lambda}}{\omega_s} = \frac{\sqrt{2} \times 216.27}{2\pi \times 60} = 0.8113 \text{ Wb}$$

- (3) For V/f control with flux level maintained to the value at rated condition, the $U_{s\lambda}$ needed at 0.1 Hz:

$$U_{s\lambda} = \frac{U_{s\lambda, \text{rated}}}{f_{s, \text{rated}}} f_s = \frac{216.27}{60} \times 0.1 = 0.36 \text{ Vrms} = 0.51 \text{ Vpk}$$

Which is 0.51 V amplitude

It is known that $i_s = 1.0 \text{ Apk}$, and $\phi_s = 30^\circ$. The amplitude of the phase voltage after voltage drop compensation can be calculated as:

$$\begin{aligned}U_s &= \sqrt{U_{s\lambda}^2 - (r_s i_s \sin \phi_s)^2} + r_s i_s \cos \phi_s \\ &= \sqrt{0.51^2 - (0.28 \times 1.0 \times \sin 30^\circ)^2} + 0.28 \times 1.0 \times \cos 30^\circ = 0.73 \text{ Vpk}\end{aligned}$$

Thus, the magnitude of phase voltage vector is 0.73 V.

- (4) The reference frequency is 50% of rated frequency, i.e. 30 Hz.

The expected rotor electrical frequency is $10 \text{ Hz} \times 3 = 30 \text{ Hz}$.

In order to make the rotor frequency equals to the reference frequency, slip compensation is expected.

It is known that: $\frac{T_e}{T_{rated}} = \frac{f_{se}}{f_{se,rated}}$.

The slip at rated condition: $s_{rated} = \frac{n_s - n_{r,rated}}{n_s} = \frac{1200 - 1160}{1200} = \frac{1}{30}$.

The slip electrical frequency: $f_{se,rated} = s_{rated} f_s = 2 \text{ Hz}$.

For 25% rated torque, the slip frequency is: $f_{se} = 0.25 f_{se,rated} = 0.5 \text{ Hz}$.

Thus, the slip frequency needed to be compensated is 0.5 Hz.